Wadim **Müller**

System Software Expert · Linux Enthusiast

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"He who stops being better stops being good." (Oliver Cromwell)

Education

Technische Hochschule Aschaffenburg (University of Applied Sciences)

Aschaffenburg, Germany

B.ENG. IN ELECTRICAL ENGINEERING AND INFORMATION TECHNOLOGY

Sep. 2008 - Aug. 2012

- Emphasis on Microelectronics with final grade of 1.5 (First Class Honours).
- Thesis: Development of an USB Device Interface based on the Test and Measurement USB Class Standard (USB TMC).

Hochschule Darmstadt (University of Applied Sciences)

Darmstadt, Germany

M.SC. IN ELECTRICAL ENGINEERING

Sep. 2014 - Aug. 2017

- Extra-occupational study with emphasis on Microelectronics with final grade of 1.1 (First Class Honours).
- Thesis: Development of an FPGA accelarated Linux System for Realtime Video Feature extraction using OpenCV on a Zynq Multicore pSoC. https://github.com/wafgo/morphological-filter-ip-core, https://github.com/wafgo/disparity-coprocessor-ip-core, https://github.com/wafgo/linux-kernel-custom-hw-uio, https://github.com/wafgo/rt-depth-map.

Skills

Programming	C, C++, Python, Perl, Bash
Virtualization/Emulation	QEMU, KVM
Debugging	Lauterbach, Trace32, GDB, OpenOCD, Greenhills MULTI, ARM Core Sight
Development Tooling	CMake, GNU make, kbuild, bitbake, gcc, binutils, llvm/clang, ghs, capstone, pyspark
Languages	German, English, Russian

Experience

CMBlu Energy AG

Alzenau, Germany

Embedded Linux Developer

since Feb. 2025

- Developed a custom Yocto Linux distribution for a Redox-Flow Battery Management System based on TI AM64xx SoC architecture.
- Performed complete board bring-up and hardware validation for custom AM64xx PCB designs.
- Designed and integrated Linux kernel drivers for input capture, PWM control, and rotary encoder interfaces.
- Implemented QEMU-based hardware emulation environment with custom kernel drivers (Counter/PWM with sysfs interfaces) to enable comprehensive application software testing and driver validation without physical hardware dependencies.

Continental Automotive GmbH

Babenhausen, Germany

Software System Integrator/Embedded Linux Developer

Jul. 2018 - Jan. 2025

- Integrated Hardware-Vendor-provided Board Support Packages into Instrument Cluster Products based on Renesas Multicore ARMv7/v8 targets running Greenhills INTEGRITY OS.
- Designed and deployed Build Systems for various Instrument Cluster Products based on CMake.
- Debugged system level errors on Multicore ARMv8/v7 Systems using JTAG with GHS (MULTI) and Lauterbach Tools (Trace32).
- Implemented several RCar3 SoC Device Emulations in QEMU. Deployed machine emulation of customized Renesas RCar3 boards using QEMU https://github.com/wafgo/qemu/tree/rcar3_port.
- Deployed a Yocto based Linux Distribution as a Flash-Frontend for Instrument Clusters running Greenhills INTEGRITY OS.
- Established a core architecture for regulating of outbound DNS traffic in multi-account and multi-VPC environments utilizing AWS Route53 DNS Firewall and FMS. This significantly increased the level of security confidence in the financial sector's segregated environment.
- Developed and Deployed Debugging Tools for GHS based systems to support low level error detection (offline stackdump unwinding on armv8, ARM ETM Setup etc.)

Magna Powertrain

Sailauf, Germany

EMBEDDED SOFTWARE ENGINEER

Jan. 2017 - Jul. 2018

- Implemented Algorithms for Brushless DC Drive Controls on resource limited Microcontrollers.
- Ported FreeRTOS to the NXP S12Z CPU Architecture.
- Designed a Software Architecture to support the development of MCU agnostic drive control algorithms including kbuild style buildsystem based on GNU make.

Rohde & Schwarz *Mainhausen, Germany*

Embedded Software Engineer *Jan. 2009 - Dec. 2016*

- Implemented several Linux Kernel Drivers for customized Freescale IMX6 based Boards used in Rohde & Schwarz Test and Measurement Devices.
- Developed Bringup Software for Freescale IMX6 and Renesas SH2A Boards.
- Developed a SPI to Ethernet bridge based on an PIC32 MCU with a MCU Firmware and the corresponding Linux Kernel Network driver on the Renesas SH7264 (SH2A).
- Developed and Deployed a Lauterbach Trace32 Debug Script Infrastructure to support early boot debugging.
- Implemented an USB Device Interface based on the TMC (Test and Measurement Standard) which is used in several Rohde & Schwarz Measurement devices for configuration and data retrieval over USB.
- Implemented many Application Level Modules (including corresponding Unit Tests) for several Test and Measurement Devices (LCR/Multimeter/H-Bridge).